

How Much of Reported Skill Is the Reference Forecast? A Controlled Evaluation for Short-Term Photovoltaic Power Forecasting

Author names withheld for double-blind review

Affiliation withheld for double-blind review

Email withheld for double-blind review

Abstract—Photovoltaic power forecasting studies report large accuracy gains under widely varying protocol: of 27 recent papers we coded, 26 report no skill score against a reference forecast and all report point estimates without run-to-run variance. Using a leakage-controlled protocol on three co-located arrays (900 recorded runs, five architectures, five seeds), we show the reference forecast determines both the magnitude and the shape of reported skill. Against smart persistence, skill rises monotonically with horizon, read as a model’s advantage growing at longer lead times; against the optimal convex combination of climatology and persistence, the same forecasts give non-monotonic skill peaking at three hours. Night hours deflate normalised error by approximately 34 percent, closed-form in the daylight fraction, while skill is almost unchanged. Reported accuracy here is shaped more by the evaluation protocol than by the model: on the test year, the reference effect (0.469) is 5.46 times the largest architectural effect we measure across all five architectures (0.086), and 16.6 times the base-architecture spread alone (0.028), both test-split figures.

Index Terms—Photovoltaic power forecasting, forecast verification, skill score, reproducibility, benchmark evaluation.

I. INTRODUCTION

Short-term photovoltaic (PV) power forecasting is an active research area, and reported accuracy improvements for hybrid deep-learning architectures are substantial. We built an open, leakage-controlled benchmark harness for this task on a public dataset: five architectures, evaluated across three module technologies, three forecast horizons, two feature regimes and five random seeds, with 900 machine-readable run records that reproduce every table and figure in this paper.

To characterise how such gains are currently reported, we coded 27 papers on hybrid PV power forecasting, published between 2022 and 2026, on eight dimensions of evaluation protocol, each judgement supported by a verbatim quotation from the source. Two counts summarise the picture: 26 of 27 report no skill score against any reference forecast, and all 27 report point estimates with no measure of run-to-run variance. Reported numbers in this literature are therefore difficult to compare with one another, or with a naive forecast, without knowing what they were measured against.

This paper measures what that comparison is worth. Holding the model, data and training procedure fixed, we vary only the reference forecast and the treatment of night hours. On one array at a six-hour horizon, on the held-out test year, the same forecasts score 0.701 against smart persistence and

0.232 against the optimal convex combination of climatology and persistence recommended by Yang et al. [1] - 67 percent of the apparent skill relative to persistence disappears when the same forecasts are evaluated against the convex reference (70 percent on the validation split, an upper bound, since the reference’s weight is fitted there). More consequentially, skill against persistence rises monotonically with horizon, the pattern usually read as a model’s advantage growing at longer lead times, while skill against the recommended reference is non-monotonic and peaks at three hours: the trend is a property of the reference, not the model. Including night hours deflates normalised RMSE by approximately 34 percent, a factor that is closed-form in the daylight fraction, while the skill score is nearly unchanged. Among the three base architectures evaluated under an identical protocol, the largest difference we measure on the held-out test year is 0.028 in skill score - smaller than either protocol effect.

A. Contributions

1. An open, leakage-controlled benchmark harness for short-term PV power forecasting, with 900 machine-readable run records reproducing every reported number.
2. A measurement of two evaluation-protocol effects - reference forecast and night-hour inclusion - held against a fixed model and data, and shown to exceed the largest architectural difference we measure.
3. A coded survey of 27 papers characterising current reporting practice on reference forecasts and run-to-run variance.

This paper is organised around three questions. RQ1: how much does the choice of reference forecast change the magnitude and shape of reported skill (Section IV-A)? RQ2: how much does including night hours change reported error and skill (Section IV-B)? RQ3: how large are these protocol effects relative to the largest difference we measure between architectures (Section IV-C)?

II. RELATED WORK

Hybrid architectures combining convolutional and recurrent layers, often with a further decomposition or gradient-boosted stage, dominate recent work on short-term PV power forecasting, and reported accuracy gains are large: a meta-analysis of 180 studies published since 2007 concludes that hybrid models consistently outperform alternatives [2]. That

TABLE I

LITERATURE SURVEY OF EVALUATION PRACTICE, $n = 27$ PAPERS ON HYBRID CONVOLUTIONAL-RECURRENT PV POWER FORECASTING, CODED ON EIGHT DIMENSIONS OF EVALUATION PROTOCOL (SECTION II); COUNTS AND PERCENTAGES ARE PER CODED VALUE WITHIN EACH DIMENSION. THE DOCUMENTED-LEAKAGE DIMENSION FOLLOWS THE TAXONOMY OF KAPOOR AND NARAYANAN [5]; THREE PAPERS IN OUR SAMPLE SHOW DOCUMENTED LEAKAGE UNDER IT [3], [6], [7]. FULL PER-PAPER CODING WITH VERBATIM SUPPORTING EVIDENCE IS AVAILABLE AT AN ANONYMISED REPOSITORY, LINK WITHHELD FOR REVIEW.

Dimension	Value	n	%
Night hours excluded	not stated	17	63.0
	yes	8	29.6
	no	1	3.7
	partial	1	3.7
Baseline used	own components only	17	63.0
	other ML models	6	22.2
	none	3	11.1
	convex combination	1	3.7
Skill score reported	no	26	96.3
	yes	1	3.7
Weather source	measured	20	74.1
	not stated	4	14.8
	NWP forecast	1	3.7
	none	1	3.7
	reanalysis	1	3.7
Split type	not stated	22	81.5
	chronological	3	11.1
	k-fold	1	3.7
	rolling-origin	1	3.7
Run-to-run variance reported	no	27	100.0
Code available	not stated	26	96.3
	yes	1	3.7
Documented leakage	none	24	88.9
	documented	3	11.1

conclusion aggregates reported errors; this paper asks what those errors are measured against.

Papers were located by keyword search for hybrid convolutional-recurrent architectures applied to photovoltaic power forecasting - search terms combining an architecture family (convolutional, recurrent, LSTM, hybrid) with photovoltaic or solar power forecasting - restricted to work published between 2022 and 2026. The inclusion criterion was a hybrid convolutional-recurrent, or closely related hybrid deep-learning, architecture evaluated for PV power output forecasting.

To characterise current practice, we coded the 27 papers meeting that criterion, on eight dimensions of evaluation protocol, each judgement supported by a verbatim quotation from the source (Table I). Beyond the two counts given in Section I, seventeen of 27 papers compare only against their own architecture's components (e.g., [3], [4]), and seventeen do not state whether night hours are excluded from evaluation - the two protocol axes this paper measures.

Rigorous verification practice exists in solar forecasting [1], [8]. Yang et al. [1], writing with more than thirty co-authors, recommend universally reporting the skill score against the optimal convex combination of climatology and persistence.

One paper in our sample of 27 meets that standard: Mayer [9] evaluates day-ahead forecasts for fourteen plants against that convex reference, states its daylight filter as a protocol, and holds out a full year used in no part of model selection; its headline hybrid improvement is 5.2 percent in mean absolute error and 1.0 percent in root-mean-square error over an optimised physical model chain, an order of magnitude smaller than improvements typically reported elsewhere in the sample, and measured against a considerably stronger baseline. The standard is achievable and is published in the same journals the surveyed papers appear in; we adopt its reference forecast here and hold the model fixed while varying it.

III. DATA AND PROTOCOL

A. Data

Experiments use the Desert Knowledge Australia Solar Centre (DKASC) dataset at Alice Springs (23.767°S, 133.867°E, 558 m), an open dataset of ground-mounted PV installations with per-array power measurement and a co-located weather station [10], also used by six of the papers in our survey [6], [11]–[15]; every number in this paper is independently checkable against it. Three fixed-mount arrays are used, chosen for differing module technology at comparable scale: array 11 (BP Solar, polycrystalline silicon, 5.0 kW), array 12 (BP Solar, monocrystalline silicon, 5.1 kW), and array 17 (Sanyo, heterojunction, 6.3 kW), all mounted at 20° tilt. The three arrays share one weather station and are not independent replications of one another.

Splits are chronological whole calendar years: training 2011–2013, validation 2014, test 2015, the last touched once, at the end. Training begins in 2011 because array 17 was commissioned in March 2010, and all three arrays share this training window so that cross-array comparisons are not confounded by differing training length.

Evaluation is restricted to hours with solar elevation above 10°, stricter than the 85° zenith filter Yang et al. [1] describe as typical practice, applied for the reason they give: at low solar elevation the clear-sky index becomes unstable relative to a small denominator. One further exclusion is applied: array 17 was switched off for a documented maintenance outage from 5 to 9 June 2015, excluded from evaluation for that array only.

B. The Excluded Array

A fourth array, array 07 (First Solar, cadmium telluride, 7.0 kW), was initially included and subsequently excluded, and is reported rather than omitted, because the mechanism by which it was nearly retained is itself evidence for this harness. A conventional completeness audit passed array 07 for 2014 with 99.99 percent coverage and 0.00 percent missing values in the power channel; the array had in fact produced no power from March to September that year, with 48.41 percent of its daylight hours recording exactly zero output. The audit did not detect this because the logger continued to record faithfully, and what it recorded was zero: zero is not missing, and a completeness-only check is structurally blind to a healthy sensor reporting a dead array. A second audit,

testing for near-zero rather than exactly-missing output, is what caught it, and both audits are included in the anonymised repository accompanying this paper.

C. Protocol

We state here only the rules bearing on this paper’s claim; the full evaluation protocol, including feature construction and significance testing, is available in the anonymised repository accompanying this paper.

Splits are never shuffled, and all scalars and feature statistics used by the models are fitted on training data only.

Three reference forecasts are used. Smart persistence forecasts $\hat{P}(t) = k_p(t - h) \times P_{cs}(t)$, persisting the clear-sky index from the issue time. Climatology forecasts $\hat{P}(t) = \bar{k}_p(\text{month}, \text{hour}) \times P_{cs}(t)$, with the mean clear-sky index per calendar month and hour of day estimated on training data only. The convex reference is $w \times \text{persistence} + (1 - w) \times \text{climatology}$, with w selected by grid search to minimise root-mean-square error on the validation split, selected independently for each array, horizon and regime; because the reference forecasts are deterministic, the weight is shared across model seeds. Because w is chosen to minimise error on that split, validation skill against the convex reference is in-sample for the reference; w is never refit on test, so the held-out test year is the clean, out-of-sample measurement, and Section IV-A reports it first. Following Yang et al. [1], the convex reference is the headline metric; skill against plain persistence is reported alongside because the gap between the two is a result, not a robustness check.

The skill score is $s = 1 - \text{RMSE}_{\text{forecast}} / \text{RMSE}_{\text{reference}}$, computed against both references over the same forecasts and the same evaluation sample.

IV. RESULTS

Figures and Table II’s array-11 configurations are for the validation split (2014) unless stated otherwise; arrays 12 and 17 are reported in the text below. The convex reference’s weight is fit on validation, so validation skill against it is in-sample for the reference (Section III-C); Section IV-A therefore reports the held-out test year (2015) first, evaluated once after every modelling and evaluation choice was frozen, as the clean, out-of-sample measurement.

A. RQ1: The Reference Forecast Changes Both Magnitude and Shape

Holding the model, the data and the training procedure fixed, we compute skill for the same forecasts against two references: smart persistence and the convex reference. The convex reference’s weight w is fitted on the validation split (Section III-C), so validation skill against it is in-sample for the reference; we report the held-out test year first, since it is the clean measurement.

On the test split (2015), evaluated once after every modelling and evaluation choice was frozen, skill against persistence for gradient boosting on array 11 is 0.266, 0.574 and 0.701 at one, three and six hours; against the convex reference,

over the same forecasts, it is 0.210, 0.316 and 0.232. The pattern holds on all three arrays: 0.263 to 0.202, 0.556 to 0.305 and 0.684 to 0.225 on array 12; 0.077 to 0.043, 0.281 to 0.156 and 0.381 to 0.144 on array 17.

The shape, not only the magnitude, changes, and the test split establishes this cleanly, out-of-sample for the reference. Measured against persistence, skill rises monotonically with horizon on every array - the pattern commonly reported as evidence that a method’s advantage grows at longer lead times. Measured against the convex reference, skill is non-monotonic on every array, peaking at three hours and falling back by six, on data no evaluation choice was tuned against.

Fig. 1 illustrates the same shape on the validation split (2014), the split on which w was fit and therefore in-sample for the reference: the solid curve (persistence) rises monotonically in every panel; the dashed curve (convex reference) peaks at three hours in every panel and falls back at six. On validation, skill against persistence for gradient boosting on array 11 is 0.252, 0.527 and 0.652; against the convex reference it is 0.200, 0.276 and 0.194 (Table II). At six hours, 70 percent of the apparent skill relative to persistence disappears when the same forecasts are evaluated against the convex reference - an upper bound, since w was fit to minimise error on this exact split. The pattern holds on all three arrays: 0.244 to 0.190, 0.510 to 0.275 and 0.639 to 0.201 on array 12; 0.166 to 0.122, 0.341 to 0.183 and 0.386 to 0.132 on array 17.

The reference effect is not a normalisation artifact. Table III gives the absolute RMSE behind the validation skill numbers above: persistence itself deteriorates sharply with horizon, from 0.428 kW at one hour to 1.548 kW at six, while climatology remains flat at 0.669 kW throughout. The denominator changes because the baseline genuinely changes in quality, not because of how the ratio is formed.

The mechanism is structural. Smart persistence forecasts $\hat{P}(t) = k_p(t - h) \times P_{cs}(t)$, and at a six-hour horizon a midday target is issued near dawn, when no valid clear-sky index yet exists: the proportion of daylight forecasts relying on a forward-filled rather than directly observed index rises from 3.3 percent at one hour to 22.6 percent at three and 51 percent at six, and the resulting bias reaches -2.15 kW at midday on a 5 kW array. The fitted convex weight w reflects this directly, falling from 0.77 at one hour to 0.25 at three and 0.04 at six on array 11 - at long horizons the optimal reference is almost entirely climatology, which has no comparable staleness problem. Array 17 is a partial exception: its fitted weight remains higher at every horizon (0.83, 0.50, 0.31), because its persistence forecast is stronger in relative terms, and its non-monotonic peak in Fig. 1 is correspondingly shallower.

Under a paired Diebold-Mariano test with HAC (Bartlett-kernel) variance and the Harvey-Leybourne-Newbold small-sample correction [16], [17], the convex reference is supported by Holm-adjusted Diebold-Mariano tests on the held-out test split, beating smart persistence in all nine array-horizon cells ($p_{\text{holm}} < 0.05$); the corresponding validation-split comparisons, reported descriptively because w is fitted to minimise

TABLE II

PROTOCOL CONFIGURATIONS, ARRAY 11, LAGGED REGIME, VALIDATION YEAR 2014: SAMPLE COUNT, NORMALISED RMSE, SKILL SCORE, AND REFERENCE FORECAST FOR SIX EVALUATION-PROTOCOL CONFIGURATIONS (C1-C6) AT EACH OF THE THREE HORIZONS. C1: DAYLIGHT HOURS ONLY, SKILL AGAINST THE CONVEX REFERENCE (THE PROTOCOL USED THROUGHOUT THIS PAPER). C2: DAYLIGHT HOURS ONLY, SKILL AGAINST SMART PERSISTENCE. C3: HOURS WHERE THE CONVEX REFERENCE HAS A PREDICTION (NOT A FULL 24-HOUR CYCLE - INCLUDED ONLY TO DOCUMENT THIS COVERAGE RESTRICTION). C4: ALL 24 HOURS, SKILL AGAINST SMART PERSISTENCE. C5 AND C6: RAW nRMSE AT DAYLIGHT-ONLY AND ALL-24-HOURS, NO REFERENCE FORECAST. COMPARING C5 AGAINST C6 SHOWS THE CLOSED-FORM NIGHT-HOUR DEFLATION; COMPARING C1 AGAINST C2 SHOWS THE REFERENCE-FORECAST EFFECT - AT $h = 6$, SKILL FALLS FROM +0.652 AGAINST PERSISTENCE TO +0.194 AGAINST THE CONVEX REFERENCE, USING THE IDENTICAL FORECASTS. RESTRICTED TO ARRAY 11; ARRAYS 12 AND 17 SHOW THE SAME PATTERN (SECTION IV).

h	Config	n	nRMSE (%)	skill	Reference
1	C1	3773	6.36	+0.200	convex
	C2	3773	6.36	+0.252	persistence
	C3	4255	6.00	+0.207	convex
	C4	8694	4.20	+0.261	persistence
	C5	3773	6.36	—	—
	C6	8694	4.20	—	—
3	C1	3762	8.97	+0.276	convex
	C2	3762	8.97	+0.526	persistence
	C3	4247	8.44	+0.276	convex
	C4	8694	5.90	+0.528	persistence
	C5	3762	8.97	—	—
	C6	8694	5.90	—	—
6	C1	3756	10.71	+0.194	convex
	C2	3756	10.71	+0.652	persistence
	C3	4245	10.07	+0.194	convex
	C4	8694	7.04	+0.652	persistence
	C5	3756	10.71	—	—
	C6	8694	7.04	—	—

TABLE III

ABSOLUTE RMSE (kW), ARRAY 11, LAGGED REGIME, VALIDATION YEAR 2014, DAYLIGHT HOURS ONLY. SKILL IS A RATIO; THIS TABLE LETS IT BE RECONSTRUCTED AND SHOWS WHETHER EACH REFERENCE IS WEAK OR STRONG IN ABSOLUTE TERMS. CLIMATOLOGY IS CONSTANT ACROSS HORIZONS BECAUSE IT USES NO RECENT OBSERVATION. PERSISTENCE IS COMPETITIVE WITH THE MODEL AT $h = 1$ BUT DEGRADES SHARPLY BY $h = 6$; CLIMATOLOGY AND THE CONVEX REFERENCE DO NOT.

h	Model	Persistence	Climatology	Convex
1	0.318	0.428	0.669	0.400
3	0.448	0.955	0.669	0.621
6	0.535	1.548	0.669	0.666

error on that exact split, are significant in all nine cells there as well. The reference recommended by Yang et al. is not merely a construction, it is a significantly harder target.

Array 17's absolute skill shifts between years by more than the other two arrays' does, and the shift traces to its reference rather than its models: array 17's convex weight leans more heavily on persistence at every horizon (0.31 to 0.83, against 0.04 to 0.77 for arrays 11 and 12), so its reference is structurally more exposed to persistence's year-to-year variation. This affects magnitude, not the shape the paper claims; the full decomposition is in the anonymised repository accompanying this paper.

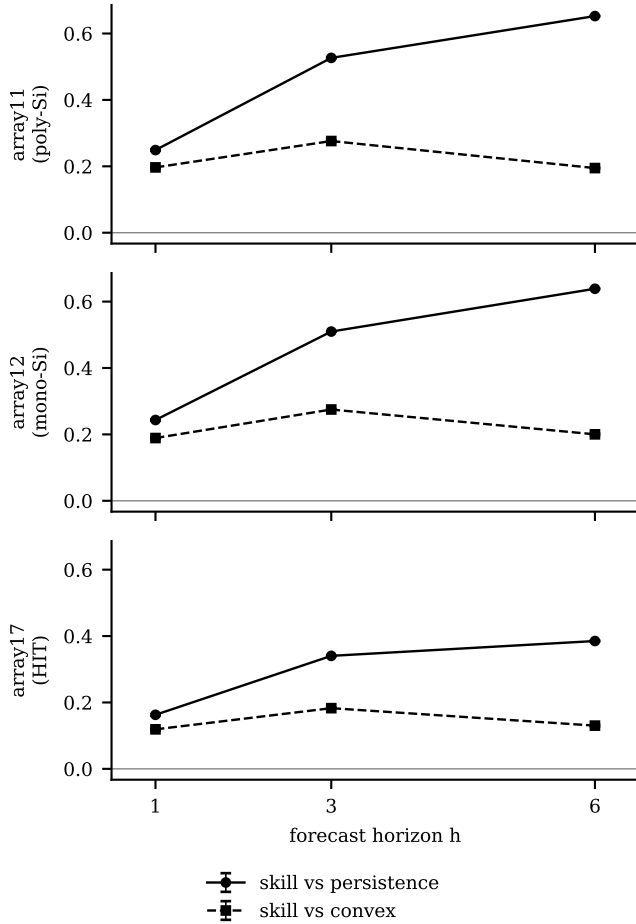


Fig. 1. Forecast skill against two reference forecasts, XGBoost, lagged regime, validation year 2014 (in-sample for the reference: w is fit to minimise error on this split), one panel per array. Skill against smart persistence (solid) rises monotonically with horizon in every panel; skill against the optimal convex combination of climatology and persistence (dashed) peaks at three hours and falls back at six - the same shape confirmed out-of-sample on the held-out test year. The apparent horizon trend is a property of the reference, not the model. Error bars show one standard deviation across five seeds and are smaller than the markers. Only XGBoost is shown; the other four architectures follow the same pattern.

B. RQ2: Night Hours Deflate Error, Not Skill

Restricting evaluation to daylight hours or retaining all 24 hours changes reported error substantially and reported skill almost not at all. If night-time errors are approximately zero, then

$$\text{RMSE}_{\text{all}} = \text{RMSE}_{\text{daylight}} \times \sqrt{N_{\text{daylight}}/N_{\text{all}}}. \quad (1)$$

At six hours on array 11, $N_{\text{daylight}} = 3756$ and $N_{\text{all}} = 8694$, giving a predicted ratio of 0.657 against an observed 0.658; agreement is within 0.3 percent at every array and horizon. Reported normalised error at this site is therefore deflated by approximately 34 percent by a protocol choice unreported in 17 of the 27 papers we surveyed (Table I), and the deflation factor is a property of latitude and season, not of the forecasting method. Skill, by contrast, is nearly invariant

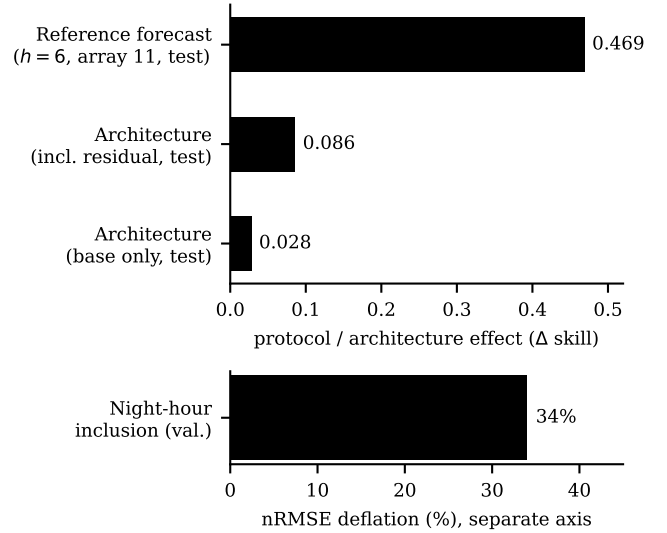


Fig. 2. Effect sizes under an identical protocol. Top axis, three bars, all on the held-out test split, same units and directly comparable: the largest difference among the three base architectures (0.028 in skill); the largest architectural effect of any kind, including the residual correction stage (0.086); and the reference-forecast effect (0.469 in skill, array 11, $h = 6$) - approximately 5.46 times the largest architectural effect of any kind. Bottom axis, validation split: the night-hour inclusion effect (34 percent nRMSE deflation), a different unit, shown on its own axis rather than a shared bar length. Under an identical benchmark, protocol effects dominate architecture effects.

to the same choice - for gradient boosting on array 11 it moves from 0.252 to 0.261, 0.527 to 0.528 and 0.652 to 0.652 at one, three and six hours - because the reference forecast is deflated by the same factor as the model.

C. RQ3: Architecture Contributes Comparatively Little

Across all five architectures we evaluate - gradient boosting, a recurrent network, a convolutional-recurrent network, and both recurrent variants with a gradient-boosted residual correction stage - the largest effect of any architectural choice we measure on the held-out test year, including the residual correction stage itself, is 0.086 in skill score. The reference-forecast effect alone at six hours on array 11, also on the test split ($0.701 - 0.232 = 0.469$ in skill), is approximately 5.46 times this largest test-split architectural effect, measured under an identical protocol. Restricted to the three base architectures only - excluding the residual correction stage - the largest test-split pairwise difference is smaller still, 0.028, making the reference effect approximately 16.6 times larger; this narrower comparison is a sub-result within the broader one above, shown in Fig. 2. The largest architectural differences on test occur at array 17 and six hours - the same array-horizon cell whose reference behaviour differs most between years, as documented in Section IV-A.

D. Held-Out Test Year

The reference-forecast shape inversion, shown above to hold on the test split, is not the only direction that does. Night hours remain deflationary and skill remains nearly unchanged by the

same closed-form argument, which does not depend on which year it is applied to, and the largest architecture difference remains an order of magnitude smaller than either protocol effect on the test split as well as on validation.

V. LIMITATIONS

All results come from three co-located arrays at a single desert site, sharing one weather station, so their forecast errors are correlated and the nine array-horizon cells are not nine independent trials. One validation year and one held-out test year are used, at horizons of one to six hours. Forecasts are deterministic; no probabilistic evaluation is performed. These results establish that the reference forecast and the treatment of night hours materially change reported accuracy at this site - not that the same magnitudes hold at a different site or climate.

VI. CONCLUSION

Holding the model, the data and the training procedure fixed, we varied only the reference forecast and the treatment of night hours. Baseline selection does not merely rescale reported skill; it can reverse the qualitative conclusion about how forecast advantage varies with horizon. The reference forecast changes both the magnitude and the shape of reported skill: on the held-out test year, 67 percent of the apparent skill relative to persistence disappears when the same forecasts are evaluated against the convex reference at a six-hour horizon (70 percent on the validation split, an upper bound, since the reference's weight is fitted there), and the trend inverts from monotonically increasing to a peak at three hours - a shape inversion established cleanly on the held-out test year, where the convex reference's weight is never fit, and supported by Holm-adjusted Diebold-Mariano tests on that same test split, with the corresponding validation comparisons reported descriptively. Including night hours deflates normalised error by approximately 34 percent, closed-form in the daylight fraction, while leaving skill nearly unchanged. Both effects exceed the largest architectural effect we measure on the test split, 0.086 in skill score, including the residual correction stage; restricted to the three base architectures alone the test-split spread is smaller still, 0.028, making the reference-forecast effect approximately 16.6 times larger in that narrower comparison and 5.46 times larger overall.

A. Recommendations

Three practices, none of them novel, would make results in this literature comparable. Report a skill score against a stated reference forecast: against smart persistence alone our results appear to improve steadily with horizon, against the convex combination of climatology and persistence recommended by Yang et al. [1] they do not, and the two references support opposite conclusions from identical forecasts. Report the evaluation sample: whether night hours are included, what daylight threshold is applied, and how many samples each cell contains are each sufficient to move a reported error by

more than the improvement typically claimed. Report run-to-run variance: none of the 27 papers we surveyed does, and a single run cannot separate a genuine effect from retraining noise.

The harness, the 900 run records, and the coded survey with its supporting evidence are available at an anonymised repository, link withheld for review.

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